

Three multimodal fusion strategies

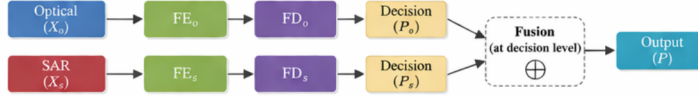
(a) Data-level Fusion



(b) Feature-level Fusion



(c) Decision-level Fusion



■ FE: Feature Encoder ■ FD: Feature Decoder ■ Decision: Task-specific Prediction

Legend

• Data-level Fusion:

The raw Optical and SAR images are fused first, and the fused data are then fed into the network for feature learning and prediction.

• Feature-level Fusion:

Each modality is encoded independently. Their deep features are fused to exploit complementary information, then decoded for prediction.

• Decision-level Fusion:

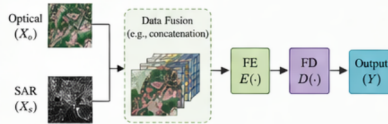
Each modality generates its own prediction, and the final result is obtained by fusing the decisions or confidence scores.

⊕: Fusion operation (e.g., concatenation / addition / attention weighting)

Detailed mechanisms of the three fusion strategies

(d) Data-level Fusion Mechanism

1) Pipeline



2) Mathematical Formulation

$$X_{fus} = \text{Concat}(X_o, X_s)$$

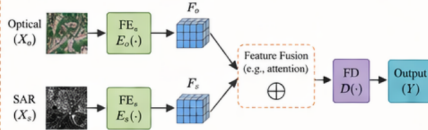
$$Y = D(E(X_{fus}))$$

3) Characteristics

- Fusion occurs at the pixel/input level.
- Preserves low-level details and complementary information.
- Sensitive to noise and radiometric differences.

(e) Feature-level Fusion Mechanism

1) Pipeline



2) Mathematical Formulation

$$F_o = E_o(X_o), \quad F_s = E_s(X_s)$$

$$F_{fus} = \alpha \odot F_o + (1 - \alpha) \odot F_s$$

$$Y = D(F_{fus})$$

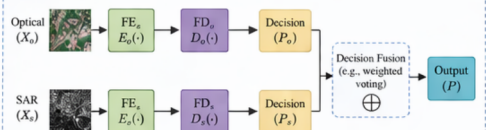
α : learnable weight
(e.g., from attention mechanism)

3) Characteristics

- Fusion occurs in the deep feature space.
- Enables high-level semantic interaction.
- Performance depends on the task, dataset, and model design.

(f) Decision-level Fusion Mechanism

1) Pipeline



2) Mathematical Formulation

$$P_o = C_o(D_o(E_o(X_o)))$$

$$P_s = C_s(D_s(E_s(X_s)))$$

$$P = \beta P_o + (1 - \beta) P_s$$

β : learnable weight
(e.g., based on confidence)

3) Characteristics

- Fusion occurs at the decision/prediction level.
- Modality-specific predictions are preserved.
- Simple and robust but may lose complementary details.

Notes: X_o, X_s : input Optical and SAR images; F_o, F_s : deep features; Y : final output (e.g., classification map / segmentation); P, P_o, P_s : prediction results.